B.Tech Project-I (CSQ401)

Project proposals — final year CSE

40 titles · 12 faculty groups · attachment to the CSQ401 allocation email

Anand M. Baswade

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- Resource optimization in LEO satellite networks – theory and/or implementation
- Dynamic spectrum sharing in wireless networks – theory and/or implementation
- Resource optimization in terahertz networks – theory and/or implementation
- Security and privacy in open radio access networks – theory and/or implementation

Anil Kumar Sao

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- Analyzing brain FC for healthy and Alzheimer's using Mamba architecture
- Open to discuss a problem on image and video processing

Amit Kumar Dhar, Dhiman Saha

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- Out-of-the-box attendance systems

Barun Gorain

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- Graph exploration by a team of mobile robots
- Distributed leader election algorithms
- Dispersion in graphs by automata
- Rendezvous by mobile robots in presence of faulty nodes in a graph

Dhiman Saha

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- High performance cryptanalysis – GPU/cluster assisted implementation of cryptanalytic attacks
- Real world cryptography – fault injection attacks
- PSK-VPN client on Android and iOS
- LLM-assisted and theoretical cryptanalysis of symmetric ciphers

Gagan Raj Gupta

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- Computer vision: noise removal ML algorithms for industrial cameras
- Computer vision: grading, chemistry and pricing of scrap metal for the steel industry
- Medical imaging: fairness and federated learning
- Medical imaging: virtual surgical planning system

Rajesh Kumar Mundotiya

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- Agentic disentanglement for controllable dialogue generation
- Disentangling agent for emotional consistency in conversational AI
- Flow network for emotion cause identification across heterogeneous modalities

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Rishi Ranjan Singh

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- DL-based network similarity
- Packing of subgraphs
- Fair influence maximization
- GNN-based network analysis

Shivam Gupta

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- Efficient algorithms for machine unlearning (theory and implementation)
- Fairness in federated client selection (theory and implementation)
- Fairness in deep learning-based healthcare models (implementation)
- Human in the machine learning loop (implementation)

Sk Subidh Ali

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- Crime detection from CCTV video feed
- Deepfake video explainability using LLM
- Vulnerable code analysis using LLM
- Document forgery detection and recovery

Vinod Reddy

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- Interval vertex coloring of graphs
- Combinatorial games on graphs
- Odd coloring of graphs
- Cops and robber games on graphs

Vishwesh Jatala

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- CSE office automation
- High performance graph LLM using GPUs